

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads to the Interstate)
Transmission System)

Docket No. RM26-4-000

COMMENTS OF PUBLIC INTEREST ORGANIZATIONS

Table of Contents

I.	Introduction	2
II.	Background	9
III.	Comments.....	9
A.	FERC has jurisdiction to regulate the interconnection of large loads to the transmission system, but must not overstep the boundaries of its jurisdiction	9
1.	The FPA provides FERC with multiple relevant sources of jurisdiction.	10
2.	FERC should continue to be mindful not to overstep its jurisdiction by regulating matters that the FPA reserves to the states.	15
B.	Studying Large Loads.....	18
1.	DOE NOPR: Principles 4 through 6.....	19
2.	Comments	20
C.	Curtable Loads.....	26
1.	DOE NOPR: Principle 7	26
2.	Comments	26
D.	Network Upgrades, Transmission Costs, and Ancillary Services	30
1.	DOE NOPR: Principles 8 through 12.....	30
2.	Comments	31
E.	Reliability Reviews.....	41
1.	DOE NOPR: Principle 14.....	41
2.	Comments	41
F.	Implementation Plans	43
1.	DOE NOPR: Principle 13.....	43
2.	Comments	43
G.	Other Issues	43
1.	Comments	43
IV.	Conclusion.....	45

I. Introduction

Pursuant to Rule 211 of the Rules of Practice and Procedure of the Federal Energy Regulatory Commission (“FERC” or “Commission”), 18 C.F.R. § 385.211, Clean Air Task Force, Inc., Coalition for Affordable Utility Services and Energy Efficiency in Pennsylvania, Earthjustice, Environmental Defense Fund, Montana Environmental Information Center, Natural Resources Defense Council, Sustainable FERC Project, Sierra Club, and Southern Environmental Law Center (jointly, the “Public Interest Organizations”) submit these comments in response to the Commission’s request for comments on the Department of Energy (“DOE” or “Department”) October 23, 2025 proposal directing FERC to initiate a rulemaking concerning the interconnection of large loads for final action by the Commission under Section 403 of the Department of Energy Organization Act¹ (“DOE Proposal” or “Proposal”).²

As noted in the ANOPR, data centers are driving an extraordinary growth in electricity demand. The North American Electric Reliability Corp (NERC) warned that the size and speed with which data centers can be connected to the grid present unique challenges for demand forecasting and system planning.³ The Commission considered

¹ 42 U.S.C. § 7173 (“DOE Organization Act”).

² Docket No. RM26-4-000, Department of Energy submits letter proposing a Proposal for final action and providing a copy of the Notice of Proposed rulemaking under RM36-6 (Oct. 7, 2025). The full text of the proposed ANOPR can be found at: <https://www.energy.gov/sites/default/files/2025-10/403%20Large%20Loads%20Letter.pdf>

³ DOE ANOPR at 1, *citing* NERC, 2024 Long- Term Reliability Assessment (Dec. 2024, updated July 2025), https://www.nerc.com/pa/RAPA/ra/Reliability%20Assessments%20DL/NERC_Long%20Term%20Reliability%20Assessment_2024.pdf at 8.

many of these challenges during its November 1, 2024, Commission-led Technical Conference Regarding Large Load Co-Located at Generating Facilities.⁴

During the conference, the Commission hosted three panel discussions with a diverse set of stakeholders, from regional grid operators (RTOs) and Independent Market Monitors to state commissioners. The conference covered many of the issues relevant to this proceeding including jurisdictional service approaches other than co-location (e.g., flexible options), impacts on reliability and resource adequacy, stability of wholesale electricity market prices, and states' concerns over affordability and consumer impacts. Subsequently, on February 20, 2025, the Commission determined that PJM Interconnection, LLC's (PJM) tariff provisions for co-location arrangements appeared to be unjust and unreasonable and initiated a show cause proceeding, which teed up the need to resolve many of the challenges raised during the technical conference and related to this proceeding.⁵

The concerns about large load interconnections all became very real in capacity auctions conducted by PJM, when its 2024 auction (for capacity years 2025/2026) resulted in capacity costs surging to an eye-popping \$14.7 billion, mostly because data centers drove prices up.⁶ PJM's most recent auction in 2025 (for capacity years

⁴ [Large Loads Co-Located at Generating Facilities, Notice of Commissioner-Led Technical Conference, AD24-11-000 \(August 2, 2024\). Following the initial notice, FERC issued three supplemental notices revising the agenda, and it issued a notice requesting comments.](#)

⁵ Order Instituting Proceedings Under Section 206 of the Federal Power Act and Consolidating With Other Proceedings, EL25-49-000 (February 20, 2025).

⁶ See PJM, 2025/2026 Base Residual Auction Report, at 3 (July 2024), <https://www.pjm.com/-/media/DotCom/markets-ops/rpm/rpm-auction-info/2025-2026/2025-2026-base-residual-auction-report.pdf>; see also Monitoring Analytics, *State of the Market Report for PJM: January through September*, at 1 (Nov. 2025), ("Data center load growth is the primary reason for recent and expected capacity market conditions, including total forecast load growth, the tight supply and demand balance, and high prices.").

2026/2027) showed similar results, with overall costs of \$16.1 billion.⁷ As PJM’s independent market monitor notes, “[d]ata center load growth is the primary reason for recent and expected capacity market conditions, including total forecast load growth, the tight supply and demand balance, and high prices.”⁸ These recent price surges in PJM illustrate how the rapid and unregulated interconnection of new large loads can unjustly and unreasonably impose extreme costs and significant reliability risks on other consumers.⁹

Capacity costs are not the only driver. With respect to transmission costs, the Union of Concerned Scientists issued a report finding that during 2024, “utilities in seven PJM states passed more than \$4.3 billion in additional costs on to customers, with billions more still . . . from local transmission upgrades made to provide transmission level service directly to data centers.”¹⁰ These costs are allocated to load-serving entities as a whole, and state regulators are responsible for allocating transmission costs amongst retail rate classes. This jurisdictional split in the cost allocation process shifts data center costs on to other customers.¹¹ Absent sufficient guardrails and information, this new

⁷ PJM, *2026/2027 Base Residual Auction Report*, at 3, <https://www.pjm.com/-/media/DotCom/markets-ops/rpm/rpm-auction-info/2026-2027/2026-2027-bra-report.pdf>.

⁸ *see also* Monitoring Analytics, *State of the Market Report for PJM: January through September*, at 1 (Nov. 2025), https://www.monitoringanalytics.com/reports/PJM_State_of_the_Market/2025/2025q3-som-pjm.pdf (“Data center load growth is the primary reason for recent and expected capacity market conditions, including total forecast load growth, the tight supply and demand balance, and high prices.”).

⁹ *See, e.g.*, Monitoring Analytics, *State of the Market Report for PJM: January through June* (Aug. 2025), at 314, https://www.monitoringanalytics.com/reports/PJM_State_of_the_Market/2025.shtml (noting that “data center load growth is the primary reason for recent and expected capacity market conditions, including total forecast load growth, the tight supply and demand balance, and high prices,” and specifically that “data center load by itself resulted in an increase in the 2025/2026 BRA revenues of \$9,332,103,858 or 174.3 percent”).

¹⁰ *Union of Concerned Scientists, Connecting Costs: Loophole Costs Customers Over \$4 Billion to Connect Data Centers to Power Grid (September 2025)*: <https://www.ucs.org/sites/default/files/2025-09/PJM%20Data%20Center%20Issue%20Brief%20-%20Sep%202025.pdf> at 1.

¹¹ *See* <https://eelp.law.harvard.edu/wp-content/uploads/2025/03/Harvard-ELI-Extracting-Profits-from-the-Public.pdf> at 14.

demand will raise power bills, increase the risk of blackouts, and create more air pollution. While PJM and other RTOs, like the Southwest Power Pool, have initiated tariff filings or stakeholder processes to integrate large load customers in the near-term,¹² DOE's ANOPR gives the Commission an opportunity to take stock of the record from these proposals and earlier proceedings to devise a comprehensive approach for interconnecting large loads in the near-term, given the widespread desire for interim options. Further, a standardized rule itself could help eliminate incentives for seeking interconnections in multiple regions, helping to create more accurate large load forecasts.

These comments present preliminary thoughts on the fourteen principles enunciated in DOE's Proposal. While DOE's Proposal articulates numerous important concepts, many of which we support, DOE has done so at a very high level. Because DOE's ANOPR is much less detailed and specific than prior notices for comparably important issues,¹³ these comments provide feedback at a similarly high level without specificity. We look forward to reviewing how FERC develops these important concepts in a NOPR in this docket, and we anticipate filing more developed and detailed comments at that time. Our views and comments may evolve as FERC develops its proposals. Generally, we believe that DOE's Proposal is correct that the rapid proliferation of new large loads presents challenges that FERC must address to maintain

¹² See, e.g., PJM Interconnection, LLC, Critical Issue Fast Path for Large Load Additions (September 25, 2025): <https://www.pjm.com/-/media/DotCom/committees-groups/cifp-lla/2025/20250915/20250915-item-07---pjm-initial-proposal-and-alternatives-considered---pjm-presentation.pdf>; Tri-State Generation and Transmission Association, Inc., 193 FERC ¶ 61,070 (2025); Southwest Power Pool, Revisions to Add the High Impact Large Load Processes and Generation Assessment, Docket No. ER26-247-000 (October 24, 2025).

¹³ For example, FERC's ANOPR that ultimately led to the issuance of Orders No. 1920 and 2023 was over 100 pages long and included detailed information about the potential need for reform and about specific reforms the agency was contemplating.

a reliable and affordable electric grid. In short, new large loads—principally data centers—are seeking to connect to the grid much faster than new generation can currently be interconnected, and the resulting imbalance in demand and supply threatens to radically increase energy prices and challenge the reliability of electric service. Because these threats to affordability and reliability are emerging rapidly, FERC should quickly address how to interconnect large loads under its FPA authority.

While FERC should act promptly to address immediate affordability and reliability challenges, FERC must also develop and enhance longer-term transmission planning and interconnection strategies to ensure that stable planning structures are sufficient to ensure reliable electric service at just, reasonable, and not unduly discriminatory rates, and are consistent with longstanding open-access principles. The Commission's goal should be to ensure that transmission-planning and interconnection procedures – for both large loads and generation – are sufficiently rigorous to realistically assess the growth in demand, anticipate changes in the generation mix, and proactively plan for the resulting transmission needs. The grid is at an inflection point, and we need to increase transmission capacity – both within and between regions – to ensure affordable, reliable electricity while we meet the moment to serve these new loads. New transmission capacity – regional and interregional, whether gained through grid-enhancing technologies, advanced conductors, or new transmission build – will bolster reliability and allow large loads to interconnect more quickly and at least cost. Connecting load centers to regions with excess supply power can reduce costs and avoid

redundant infrastructure by allowing regions to share their generating resources.¹⁴ This infrastructure can also save customers money - \$7.8 billion to \$10.6 billion annually according to a recent study of interregional transmission expansion recommended by NERC.¹⁵ Further, significant capacity gains can be made fairly quickly through the addition of advanced transmission technologies (ATTs) to the system. ATTs can boost capacity by an additional 20 to 100 GW of peak demand without building new wires.¹⁶ In this rulemaking and other dockets, FERC should consider ways within its authority to encourage the adoption of these technologies as an additional way to meet growing loads.

In developing strategies to address the rapid proliferation of new large loads, whether in the short term or the long term, the Commission should follow the guiding principle that new large loads should not be permitted to harm affordability or reliability for existing consumers. Consistent with the Federal Power Act's (FPA) fundamental cost-causation principles, new large loads should be responsible for the costs they cause, and it would be unjust and unreasonable for the costs caused by new large loads to be shifted to existing consumers who did not cause those costs and who receive no, or trivial, benefits from the infrastructure developed to serve new large loads. Similarly, new large loads should not imperil reliability for other consumers; for example, it would not be just and

¹⁴ Audun Botterud et al., Bridging the Gaps: The Impact of Interregional Transmission on Emissions and Reliability, MIT Center for Energy and Environmental Policy Research, August 2024, <https://ceepr.mit.edu/workingpaper/bridging-the-gaps-the-impact-of-interregional-transmission-on-emissions-and-reliability/>.

¹⁵ Michael Goggin, Zach Zimmerman, and Dana Ammann, NERC's Recommended Grid Expansion Would Save Consumers Billions, Grid Strategies and NRDC, February 2025, https://gridstrategiesllc.com/wp-content/uploads/GS_NRDC_NERCs-Recommended-Grid-Expansion-Report54.pdf.

¹⁶ See ESIG, Grid-Enhancing Technologies: Priorities for a Modern, Efficient, and Flexible Grid 8–9 (July 2025), available at <https://www.esig.energy/wp-content/uploads/2025/07/ESIG-Grid-Enhancing-Technologies-report-2025.pdf>;

reasonable for a data-center boom to cause residential or industrial ratepayers to experience increased risks of blackouts. Finally, any new rules adopted by FERC cannot unduly preference large load customers over other types of customers.

Under the DOE Organization Act, FERC has exclusive jurisdiction over the DOE Proposal.¹⁷ After reviewing the comments it receives in response to its notice, FERC may:

- (1) concur in adoption of the rule or statement as proposed by the Secretary;
- (2) concur in adoption of the rule or statement only with such changes as it may recommend; or
- (3) recommend that the rule or statement not be adopted.^[18]

The Secretary's letter specifically anticipates that "[the Commissioners] and the Commission's dedicated staff will build upon these principles and work expeditiously to initiate a rulemaking proceeding and ultimately issue a final rule." If FERC adopts DOE's proposal, FERC must provide clear regulations to implement large load interconnection, consistent with its statutory obligation under the FPA that the rates, terms, and conditions of service must be just, reasonable, and not unduly discriminatory. We look forward to seeing how FERC develops these ideas into a more comprehensive proposal.

Importantly, large load interconnection affects a variety of energy users and providers, including Tribes. We recommend that FERC consult Tribes pursuant to the federal government's trust responsibilities.¹⁹

¹⁷ 42 U.S.C. § 7173 ("DOE Organization Act").

¹⁸ 42 U.S.C. § 7174 ("DOE Organization Act").

¹⁹ See Alliance for Tribal Clean Energy Initial Comments in Docket No. RM26-4-000; See also *Seminole Nation v. United States*, 316 U.S. 286, 296–97 (1942); *Eric v. Sec'y of U. S. Dep't of Hous. & Urban Dev.*, 464 F. Supp. 44 (D. Alaska 1978).

II. Background

On October 23, 2025, the Secretary of Energy signed an advance notice of proposed rulemaking entitled Ensuring the Timely and Orderly Interconnection of Large Loads. The Secretary proposed the rule under Section 403 of the DOE Organization Act, which allows the Department to propose rules, and reasonable timelines for final action on those rules, for consideration by FERC. The Proposal states that in light of unprecedented growth of large loads, FERC should standardize interconnection procedures and agreements for such loads, including those seeking to share a point of interconnection with new or existing generation facilities (co-located loads).²⁰ DOE includes a set of fourteen principles it believes should inform FERC's rulemaking on this issue.

III. Comments

A. FERC has jurisdiction to regulate the interconnection of large loads to the transmission system, but must not overstep the boundaries of its jurisdiction

DOE asserts that the FPA allows the Commission to exercise jurisdiction over the interconnection of large loads. First, DOE states that large load interconnections are a "critical component of open access transmission service" akin to generator interconnections that require minimum terms and conditions to ensure non-discriminatory

²⁰ PIOs note that the DOE ANOPR called these loads "hybrid facilities". However, in our comments, we use the term co-located loads as used in previous FERC orders. FERC recently adopted PJM's definition of co-located loads "as a configuration through which end-use customer load is physically connected to the facilities of an existing or planned generation unit on the interconnection customer's side of the point of interconnection to the ... transmission system." *See* PJM Interconnection, LLC, 190 FERC ¶ 61,115 (2025) at P3. We do not necessarily advocate that FERC must continue to use this definition, but use it for continuity in these comments.

transmission service.²¹ Second, DOE states that the interconnection of large loads is a practice directly affecting FERC-jurisdictional wholesale electricity rates. Finally, DOE states that its proposal does not impinge on States' authority over retail electricity sales because it does not exert jurisdiction over any retail sales to the large load. Principle 1 of the ANOPR explicitly provides that to avoid even arguably affecting the States' jurisdiction, the Commission's jurisdiction over large load interconnections should be limited to interconnections directly to transmission facilities, consistent with the Commission's seven-factor test.²²

1. The FPA provides FERC with multiple relevant sources of jurisdiction.

The FPA grants the Commission several sources of jurisdiction that are relevant to the regulation of new large load interconnection, including authority over interstate transmission, the responsibility to ensure just and reasonable and not unduly discriminatory wholesale rates by regulating practices that directly affect rates in wholesale energy markets, and responsibility to ensure electric reliability. As the Commission works to develop the principles DOE has suggested, it should ground its proposed rule in these several sources of authority.

Large loads exist at the intersection of state and federal jurisdiction. DOE's ANOPR makes clear that FERC should avoid even arguably affecting the States' jurisdiction. Thus, as the Commission considers how to proceed on regulating the interconnection of large loads, it must consider whether its authority presents a sufficient

²¹ DOE ANOPR at P 13, citing Order No. 2003 at P 9, 11.

²² DOE ANOPR at P 18.

jurisdictional foundation to adopt each of the principles that DOE suggests. We urge FERC to work cooperatively with its state colleagues to ensure that federal and state rules work seamlessly to interconnect these loads in a manner that prioritizes the protection of other consumers from unjust, unreasonable, and unduly discriminatory rates and reliability issues.

Section 206 of the FPA provides FERC with the authority and responsibility to ensure that the rates and practices within its jurisdiction are just and reasonable.²³ As discussed throughout our comments, the interconnection of large loads can shift costs to other consumers in an unjust and unreasonable manner. It is incumbent upon FERC to adopt rules sufficient to ensure that rates subject to FERC's jurisdiction will be just and reasonable moving forward. Here, this statutory responsibility means that the Commission must ensure that the interconnection of new large loads will no longer impose unjust and unreasonable expenses on other ratepayers.

²³ 16 U.S.C. 824e.

a. FERC’s jurisdiction over interstate transmission

The FPA authorizes the Commission to regulate “the transmission of electric energy in interstate commerce.”²⁴ The FPA prohibits “unreasonable rates and undue discrimination ‘with respect to any transmission or sale subject to the jurisdiction of the Commission’” and gives FERC the authority and responsibility to remedy the unlawful rates and discriminatory practices.²⁵ The FPA’s grant of jurisdiction over interstate transmission provides clear authority for the Commission to regulate interconnections to the interstate transmission system.²⁶ DOE’s Proposal suggests that the Commission should limit any rule in this docket to regulating solely connections to the interstate transmission system, as defined by FERC’s seven-factor test for distinguishing transmission infrastructure from distribution infrastructure. PIOs agree that regulating interconnection to the transmission grid, as opposed to interconnection to the distribution grid, falls within the Commission’s jurisdiction. Accordingly, we support DOE’s identification of the Commission’s jurisdiction over transmission as one foundation for this rulemaking.

b. FERC’s jurisdiction over practices directly affecting wholesale electricity rates

In addition to FERC’s jurisdiction over the transmission of electric energy in interstate commerce, the FPA also authorizes FERC to regulate “the sale of electric energy at wholesale in interstate commerce,” which includes the authority to regulate

²⁴ *FERC v. EPSA*, 577 U.S. 260, 266, 277 (2016).

²⁵ *New York v. FERC*, 535 U.S. 1, 7 (2002).

²⁶ *See Nat’l Ass’n of Reg. Util. Comm’rs v. FERC*, 475 F. 3d 1227, 1279-80 (D.C. Cir. 2007).

wholesale electricity rates and any rule or practice directly affecting such rates.²⁷ FERC must ensure that “[a]ll rates and charges made, demanded, or received by any public utility for or in connection with the transmission or sale of electric energy subject to the jurisdiction of the Commission shall be just and reasonable,”²⁸ and FERC has the authority to remedy any unreasonable or unduly discriminatory practices.²⁹ This jurisdiction gives FERC the authority to correct unjust and unreasonable rates or unduly discriminatory or preferential practices that may arise with the interconnection of large loads.

The rapid interconnection of new large loads to the transmission system—and the terms on which those new large loads are incorporated into FERC-jurisdictional markets—are practices directly affecting wholesale rates, terms and conditions for energy, ancillary services, and capacity market prices. For example, the incorporation of new large loads as demand in PJM’s capacity market is one of the principal drivers of recent dramatic increases in capacity market clearing prices. PJM identified a “3,243 MW increase in forecasted load” as one of the drivers of skyrocketing prices in the 2025/2026 Base Residual Auction (“BRA”).³⁰ Likewise, PJM identified a “5,446.1 MW increase in forecasted load” as one of the “[k]ey planning parameter changes” that led the capacity market to clear at its cap in the 2026/2027 BRA.³¹ And PJM expects this trend to

²⁷ *EPSA*, 577 U.S. at 277.

²⁸ 16 U.S. Code § 824d(a).

²⁹ *New York*, 535 U.S. at 7.

³⁰ PJM, *2025/2026 Base Residual Auction Report*, at 3, (July 30, 2024), <https://www.pjm.com/-/media/DotCom/markets-ops/rpm/rpm-auction-info/2025-2026/2025-2026-base-residual-auction-report.pdf>.

³¹ PJM, *2026/2027 Base Residual Auction Report*, at 3 (July 22, 2025), <https://www.pjm.com/-/media/DotCom/markets-ops/rpm/rpm-auction-info/2026-2027/2026-2027-bra-report.pdf>.

continue at least through the 2030s. PJM’s Independent Market Monitor (“IMM”) similarly explains that “the failure to recognize and address the role of large data center loads is a direct cause of higher prices and will continue to result in even higher prices unless the related issues are addressed.”³² Because the rules for how to incorporate new large loads into FERC-jurisdictional markets such as PJM’s capacity market are practices directly affecting wholesale rates, the Commission “has the authority—and, indeed, the duty—to ensure that rules or practices ‘affecting’ wholesale rates are just and reasonable.”³³ As the Commission develops DOE’s Proposal into a regulation, it should take care to ensure that new large loads do not cause unjust and unreasonable costs for other ratepayers.

c. FERC’s authority under section 215 of the FPA

FERC also has jurisdiction to regulate the interconnection of large loads under FPA section 215. Section 215 of the FPA outlines FERC’s reliability jurisdiction, which could serve as an avenue for FERC to address the reliability concerns associated with the interconnection of incoming large loads. Section 215 of the FPA gives FERC jurisdiction over “all users, owners and operators of the bulk-power system,” and mandates that “[a]ll users, owners and operators of the bulk-power system shall comply with reliability standards.”³⁴ A large load interconnected to the transmission system would clearly be a user of the bulk-power system. Section 215(d)(5) also allows FERC to direct the North

³² Monitoring Analytics, *State of the Market Report for PJM*, at 1, 306 (Aug. 14, 2025), https://www.monitoringanalytics.com/reports/PJM_State_of_the_Market/2025.shtml

³³ *EPSA*, 577 U.S. at 266, 277.

³⁴ 16 U.S.C. § 824o(b)(1).

American Electric Reliability Corp, which develops reliability standards subject to FERC oversight, to develop or modify standards to address a specific matter.³⁵ NERC is currently evaluating the effects of large loads on the reliability of the bulk-power system and whether it needs to act to set standards for large loads. If necessary, FERC could order NERC to submit new or modified reliability standards to FERC to address the reliability of large loads.

2. FERC should continue to be mindful not to overstep its jurisdiction by regulating matters that the FPA reserves to the states.

While it is important for FERC to exercise its jurisdiction over some issues affecting the interconnection of large loads and their imposition of costs on other consumers—especially to regulate issues like wholesale market practices where states lack authority—it is also important for FERC to recognize that the states have authority here as well. As noted in the DOE ANOPR, FERC must take care not to “even arguably affect[] the [s]tates' jurisdiction over generation facilities, facilities used in local distribution or only for the transmission of electric energy in intrastate commerce, or transmissions consumed by the transmitter.”³⁶ While FERC has the authority to regulate wholesale sales of electricity in interstate commerce and the transmission of electric energy in interstate commerce, the law “leaves to the States alone the regulation of ‘any other sale,’ maintaining ‘a zone of exclusive state jurisdiction,’”³⁷

³⁵ 16 U.S.C. § 824o(d)(5).

³⁶ DOE ANOPR at P 18.

³⁷ *EPSA* at 280.

States have an interest in ensuring resource adequacy and protecting consumers from cost shifts. Many states have adopted policies that can make it easier to connect new large loads while also addressing the affordability and reliability of electricity to other customers.³⁸ Some include measures like requiring a new large load to subsidize energy efficiency efforts in the local area to free up capacity for the large load while benefiting local residents and businesses or providing rate relief to certain retail customers. For example, a recently approved settlement requires three large data center customers to contribute \$500,000 annually for five years towards the Indiana Community Action Association to be used for low-income assistance, weatherization, and other programs.³⁹ Other policies seek to grant queue priority for large customers who benefit state economic development priorities or communities, including the recently approved Evergy Kansas settlement and the proposed changes to ComEd's GT&C tariff in Illinois.⁴⁰ Importantly, many state utility commissions have either already approved or are in the process of establishing large load tariffs intended to insulate other customers from inappropriately bearing the costs utilities incur to interconnect large loads. FERC must ensure that any rules it adopts do not impede these programs.

³⁸ See generally, Andrew Satchwell et al., Lawrence Berkeley Laboratory, *Electricity Rate Designs for Large Loads: Evolving Practices and Opportunities (January 2025)* at 3-4, 10-11 (describing the types of controls that states are implementing).

³⁹ See Indiana Utility Regulatory Commission Cause No. 46097 (<https://iurc.portal.in.gov/docketed-case-details/?id=b8cd5780-0546-ef11-8409-001dd803817e>).

⁴⁰ See Kansas Corporation Commission Docket No. 25-EKME-315-TAR (<https://estar.kcc.ks.gov/estar/portal/ksc/PSC/DocketDetails.aspx?DocketId=af2fbd6a-5ee7-4ad1-a3ae-4dfff1afcb87>), and Illinois Commerce Commission Docket No. 25-0680 (<https://icc.illinois.gov/docket/P2025-0680/documents>).

FERC recently recognized that its jurisdiction over large loads is limited. On October 27, 2025, FERC rejected Tri-State Generation and Transmission Association (Tri-State)’s proposed High Impact Load (HIL) Tariff.⁴¹ Tri-State, a nonprofit wholesale generation and transmission cooperative, filed a proposed tariff on August 28, 2025, which included a comprehensive study process to evaluate High Impact Load requests (the “HIL program”).⁴²

FERC correctly decided that that Tri-State’s proposal presented “an impermissible intrusion on retail rate regulation and thus [was] not subject to the Commission’s authority pursuant to section 205 of the FPA.”⁴³ FERC found that Tri-State’s HIL Tariff dictated specific terms and conditions of the retail sale of electricity, specifically by setting requirements for certain minimum load and energy amounts and maximum monthly retail energy or demand caps.⁴⁴

FERC also reasonably declined to address other jurisdictional issues raised by the protesters, particularly questions regarding the extent of FERC’s jurisdiction over transmission-level interconnections of large loads, as that issue was not properly presented in that docket. However, DOE’s Proposal may now require FERC to grapple with broader questions about the extent of its jurisdiction over the interconnections of large loads to the transmission system.

⁴¹ 193 FERC ¶ 61,070, at P 1 (2025).

⁴² *Id.*

⁴³ *Id.* at 19.

⁴⁴ *Id.* at 21.

The electric system is complex and sits at the confluence of state and federal jurisdiction, and FERC and the states have had to work cooperatively on many issues. As the Supreme Court has noted, the FPA incorporates principles of “cooperative federalism” with respect to the roles of state and federal regulators in energy regulation.⁴⁵ For example, the Court has acknowledged the interconnectedness of wholesale and retail ratemaking, holding in *Hughes v. Talen Energy Marketing*, that “[s]tates, of course, may regulate within the domain Congress assigned to them even when their laws incidentally affect areas within FERC’s domain,”⁴⁶ while holding in *FERC v. Electric Power Supply Association* that “a FERC regulation does not run afoul of [the FPA] just because it affects—even substantially—the quantity or terms of retail sales.”⁴⁷ The Court further established that the FPA “makes federal and state powers ‘complementary’ and ‘comprehensive,’ so that ‘there [will] be no “gaps” for private interests to subvert the public welfare.’”⁴⁸

Following with the spirit of the cooperative federalism principles outlined in *EPSA* and *Hughes*, we encourage FERC to set minimum standards governing the interconnection of new large loads—floors, not ceilings—to allow for more rigorous state regulation and encourage RTO/ISOs and utilities to work closely with state regulators to address the incoming large load demands.

B. Studying Large Loads

⁴⁵ *EPSA*, 577 U.S. at 288.

⁴⁶ *Hughes*, 578 U.S. at 164.

⁴⁷ *EPSA*, 577 U.S. at 281.

⁴⁸ *Id.* at 289.

1. DOE NOPR: Principles 4 through 6

DOE set out several principles that concern how the FERC rule should address studying large loads. Principle 3 provides that, “to the extent practicable, load and [co-located loads] should be studied together with generating facilities.”⁴⁹ DOE states that this will allow for efficient siting of loads and generating facilities, minimizing the need for costly network upgrades. The ANOPR provides an example of siting a large load near or at the same point of interconnection as a new generating facility, which could reduce the network upgrades needed to interconnect only the load or only the generating facility.

Principle 4 would also require large loads and co-located loads to be subject to standardized study deposits, readiness requirements, and withdrawal penalties.⁵⁰ According to DOE, “[t]hese provisions deter speculative projects and provide transmission providers with more useful information to more accurately forecast demand on their systems.” DOE also explicitly seeks comment on whether additional commitments or financial penalties would be appropriate.

Principle 5 provides that co-located loads should be studied based on the amount of injection and/or withdrawal rights requested as an incentive for large loads to co-locate with new generation. DOE provides an example of a co-located load consisting of a 500 MW load and a 600 MW generating facility seeking no withdrawal rights and 100MW of injection rights.

⁴⁹ DOE ANOPR at P 20.

⁵⁰ DOE ANOPR at P21.

Principle 6 provides that “any [co-located load] interconnection shall be required to install the system protection facilities necessary to prevent unauthorized injections or withdrawals that exceed the respective rights.”

2. Comments

We agree with the principles in the DOE proposal that seek to maintain reliability and minimize cost while interconnecting large loads. However, this must be paired with a principle that new large loads cannot disadvantage existing customers – either from a reliability or cost perspective. Further, as discussed above, we believe that FERC should look at this as a near-term issue, while also looking for a long-term solution that will create transmission planning, interconnection, and market processes that can successfully plan for the integration of these large loads as a regular course of business. We understand that we are in a period of rapid integration of large loads to the grid while it takes time to plan and build new generation and transmission resources, and FERC may need to address this near-term need with specialized tools. If we can get transmission planning and interconnection rules right, together with proper forecasting, grid operators can mitigate the issues in the long-term. This must include consideration of the interregional transmission capacity that is needed to ensure reliability, which will also help integrate new loads.⁵¹ Importantly, in the long term, building out the system in response to discrete, emergent needs rather than through proactive, forward-looking, system-wide planning fails to identify the most cost-effective infrastructure investments,

⁵¹ See generally NERC Interregional Transfer Capability Study, *available at* <https://www.nerc.com/initiatives/additional-initiatives/interregional-transfer-capability-study-its>.

which would solve the reliability issues associated with new large loads *and* provide broader system benefits.⁵² The magnitude of estimated load growth necessitates a more systematic approach to addressing these needs in the long term to ensure a reliable and affordable grid.

To that end, Principle 3 is a step in the right direction. Studying system needs holistically is essential to ensuring least-regrets system planning. FERC should set a long-term course where utilities plan for their system needs - transmission, generator interconnection, and the needs of loads – by either more closely coordinating existing transmission planning and generator interconnection processes or in a single comprehensive process. Accounting for all system needs by considering transmission planning, generator interconnection, and the need to serve all loads will ultimately increase transparency, reduce costs, and cut inefficiencies.⁵³ That said, we recognize the need for processes to quickly interconnect new loads in a just and reasonable and not unduly discriminatory way while the Commission and states work on a long-term solution.

⁵² See Order No. 1920 at P 136 (“By expanding the transmission planning horizon and considering factors affecting Long Term Transmission Needs, as well as considering a broader list of benefits, transmission providers will be able to identify, evaluate, and select more efficient or cost-effective transmission solutions to address Long-Term Transmission Needs. Such Long-Term Regional Transmission Planning will: (1) reduce reliance on transmission solutions that are relatively inefficient or less cost-effective because they address only short-term transmission needs; (2) unlock the benefits of economies of scale in transmission investment; (3) enable opportunities to “right size” replacement transmission facilities; (4) facilitate the selection of regional transmission facilities that could address multiple transmission needs over different time horizons; and (5) provide states, utilities, customers, and other stakeholders with greater insight and transparency into the costs and benefits of particular transmission solutions to address Long-Term Transmission Needs. We conclude that these regional transmission planning and cost allocation reforms will benefit customers by leading to more efficient or cost-effective transmission investment, thereby helping to ensure just and reasonable rates”). See also PIO Comments (Docket No. EL25-49-000) at 13-14.

⁵³ See SPP’s CPP https://elibrary.ferc.gov/eLibrary/filelist?accession_number=20251103-5149&optimized=false&sid=7c8ec448-abb4-495d-b95e-f5d1266d32e4

In setting up a short-term solution structure, FERC must ensure that any new processes do not disadvantage current market participants or interconnection customers. There are at least three ways a large load can acquire its own generation: (1) enter into an agreement with existing generation for all or most of its energy production, (2) enter into an agreement with a generator in the current interconnection queue, or (3) develop its own new generation. In addition, FERC should make clear that these studies can and should include studying battery storage – either co-located with the large load or as an addition to existing generation - as an option for integrating large loads.⁵⁴ Battery storage can be deployed quickly—on average within 1.7 years—making them an ideal part of the solution to the short-term concerns about interconnecting new large loads.⁵⁵ Moreover, during much of the year, there is sufficient capacity to serve this new demand. It is only during a few hours and days when demand would exceed supply that additional generation would be needed. Battery storage is a good solution to meet this short-term need. The Nicholas institute found that the grid could integrate 98 GW of new load at an average annual load curtailment rate of 0.5%, and 126 GW at a rate of 1.0%⁵⁶ - some of that capacity could be addressed by the addition of battery storage.

⁵⁴ While outside the scope of this rulemaking proceeding, PIOs also note that significant capacity gains can be made through the addition of Advanced Transmission Technologies (ATTs) to the system. ATTs can boost capacity by an additional 20 to 100 GW of peak demand without building new wires. FERC should consider ways within its authority to encourage the adoption of these technologies as an additional way to meet growing loads. See ESIG, *Grid-Enhancing Technologies: Priorities for a Modern, Efficient, and Flexible Grid* 8–9 (July 2025), available at <https://www.esig.energy/wp-content/uploads/2025/07/ESIG-Grid-Enhancing-Technologies-report-2025.pdf>;

⁵⁵ See Solar Energy Industries Association, *Why Batteries Are the Electric Grid's Most Powerful Asset*, (Apr. 2025), <https://seia.org/blog/why-batteries-are-the-electric-grids-most-powerful-asset/> ("a new battery project takes approximately 20 months on average").

⁵⁶ Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in US Power Systems at 2, available at <https://nicholasinstitute.duke.edu/sites/default/files/publications/rethinking-load-growth.pdf>

As discussed below with respect to Principle 10, FERC should not allow any single customer to claim either full or partial rights to the generation from an existing generator because it can cause reliability and resource adequacy concerns for the rest of the system.

Still, we support the concept of studying load with generation to minimize the need for network upgrades. Notably, any FERC rule must ensure that studying generation with only these large loads does not disadvantage current market participants or interconnection customers. It already takes far too long to interconnect a generator to the grid. Adding one-off processes can add administrative and modeling complexity to utility processes that may delay the utility's timely processing of existing queue requests, reduce transparency, and further erode stakeholder confidence in the interconnection process. FERC should emphasize accelerating interconnection processes generally over adopting separate fast-tracks, and must at least implement safeguards to prevent fast-tracks from causing harm to the existing generator interconnection process.

Moreover, FERC must ensure that any decision to study generation with large loads does not create unintended consequences. Using DOE's example of siting a large load near or at the same point of interconnection as a new generating facility – FERC must ensure that such a choice does not force network upgrade costs that would otherwise be shared onto other customers or degrade reliability or resource adequacy for others. For example, if that generator is already in the queue, it may have been planned to address a resource adequacy concern in the region. FERC should not allow the new large load to automatically assume that the generator will serve only that load without assessing its

effects on the region. In addition, removing a generator from the queue could have negative impacts on the rest of the queue – particularly if it has already been assigned to a cluster or been partially studied. This could lead to re-studies or impose costs on other generators in the queue in an unjust and unreasonable way that is inconsistent with Order No. 2023. In creating new rules, FERC should set out safeguards that ensure that any decision to pair a large load with new generation does not impact existing customers or other generators in the queue.

We generally agree with Principle 5 that co-located loads should be studied based on the amount of injection and/or withdrawal rights requested. That said, such a study must be paired with a commitment not to take service from the grid that exceeds that which was studied. In DOE's example of a co-located loads consisting of a 500 MW load and a 600 MW generating facility seeking no withdrawal rights and 100MW of injection rights, that means that the load should not be allowed to take transmission above that 100 MW unless it is available after all other customers have taken service. We discuss curtailment principles below.

Moreover, DOE's Principle 5 does not address what happens if the generator is on outage or operating below 500 MW output. In that case, the large load's right to take transmission service should be reduced in parallel. In sum, unless appropriate control technologies are included, as DOE's Principle 6 contemplates, or the load is willing to completely curtail, it should have to be studied at whatever minimum level it wants to be guaranteed to operate at, and also have withdrawal rights for that amount.

Principle 6 in DOE’s proposal offers a common-sense way to ensure that co-located loads remain within the limits of their agreed-upon terms of service. Principle 6—that “any hybrid interconnection shall be required to install the system protection facilities necessary to prevent unauthorized injections or withdrawals that exceed the respective rights”—appropriately focuses on control technologies as a primary approach rather than on financial incentives. Control technologies are a better approach because they can allow direct control of injections or withdrawals and thus prevent any harm associated with unauthorized actions, such as distortions in market prices or threats to reliability. While penalties for unauthorized injections or withdrawals may also be helpful, they will not prevent unauthorized actions or their harmful consequences in the first instance. Moreover, unless penalties are set high enough, an entity that stands to gain more from an unauthorized withdrawal or injection than it would lose from the penalty may simply view the penalty as the cost of doing business—and may unjustly and unreasonably impose excess costs or reliability risks on other consumers. Control technologies are a superior approach, and we encourage FERC to consider requiring the use of appropriate control technologies to control withdrawals for all new large loads rather than only for co-located facilities.

Finally, FERC should consider whether the rules should provide a pathway to firm service by either requiring the curtailable interconnection to be some form of temporary authorization, with an obligation to study what would be necessary to fully serve the load at some defined point in the future, or allow a large load to request a study on what would be necessary for full service in the future.

With respect to Principle 4, we agree that FERC should set minimum standards to promote accurate load forecasting and guard against the risk of speculative or duplicative large load interconnection requests. We agree that FERC could use the standards set in Order No. 2023 and the NERC July 2025 Large Loads Taskforce Whitepaper⁵⁷ as a guide to setting load interconnection standards, but reserve comments on this until we see specific proposals. Regions could go above and beyond these minimal standards, but FERC should establish a floor.

C. Curtailable Loads

1. DOE NOPR: Principle 7

Principle 7 states that large loads and/or co-located loads that agree to be “curtailable and dispatchable” should be eligible for expedited interconnection. The ANOPR specifically seeks comment on details of any such expedited interconnection process.

2. Comments

We agree with the overarching reality that demand from new large loads threatens to outstrip supply, especially during grid emergencies and extreme weather events. This threat increases the risk of blackouts, mandatory curtailments, and high scarcity prices across the United States for all customers. As such, it is important that both FERC and states grapple with questions of curtailment priority and flexibility requirements for new large loads to ensure that this risk is appropriately allocated, while respecting

⁵⁷ <https://www.nerc.com/globalassets/who-we-are/standing-committees/rstc/whitepaper-characteristics-and-risks-of-emerging-large-loads.pdf>

jurisdictional boundaries. Significant gains can be made with fairly minor curtailment. According to the Nicholas Institute, “76 GW of new load—equivalent to 10% of the nation’s current aggregate peak demand—could be integrated with an average annual load curtailment rate of 0.25%.”⁵⁸ Increasing that to 0.5% could yield 98 GW of new load and 1% would lead to 126 GW of new loads. Moreover, Duke and the Nicholas Institute recently noted that large load flexibility can achieve three key benefits:

- (1) faster interconnection times for large loads,
- (2) avoided or deferred transmission investments, and
- (3) avoided generation capacity investments, and/or near-term reliability and affordability for all customers as supply catches up with rising demand.⁵⁹

Duke/Nicholas recognize that DOE stated its curtailment principle at a high level and leaves to FERC the question of how to implement a large load interconnection rule that realizes the benefits of flexibility. They lay out four issues that FERC would need to address, which we endorse:

1. Enable large load flexibility commitments to avoid or defer transmission upgrades
2. Enable large load flexibility commitments to reduce capacity procurements
3. Enable flexible large loads to access non-firm withdrawal and, as applicable, injection rights
4. Clarify study timelines, queue priority, and interconnection process for large load customers, as well as load-supply [co-located loads] facilities and pairings.⁶⁰

⁵⁸ Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in US Power Systems at 2, *available at* <https://nicholasinstitute.duke.edu/sites/default/files/publications/rethinking-load-growth.pdf>

⁵⁹ Duke/Nicholas Institute, How DOE’s Proposed Large Load Interconnection Process Could Unlock the Benefits of Load Flexibility Considerations for Rulemaking at 3, *available at* <https://nicholasinstitute.duke.edu/sites/default/files/publications/how-does-proposed-large-load-interconnection-process-could-unlock-benefits-load-flexibility.pdf>.

⁶⁰ *Id.*

Large load interconnection can and should be managed via curtailment until new generation not already in the queue is added. Before speaking to the details of any expedited interconnection priority for large loads that “agree to be curtailable and dispatchable,” we believe it is crucial to first establish four principles that should guide any decision regarding the curtailment or expedited interconnection of large loads. This has been a key issue in a recent PJM “Critical Issue Fast Path” (CIFP) process on the integration of large loads, and we will use examples from the PJM CIFP process where appropriate.

First, FERC must not infringe upon state retail authority. Second, FERC must establish guidelines to limit unfair allocation of reliability risks at the wholesale level. Third, FERC should establish guidelines that balance two competing risks: the risk of curtailment on large loads, and the risk of high energy prices or scarcity prices on all customers. Curtailed too early, and large loads face potentially long and frequent outage periods, during which they may need to rely on backup generation. Curtailed too late, and all other customers will be forced to shoulder high energy prices and even scarcity prices that can reach millions of dollars per minute. Striking the right balance to ensure fair service is provided to large loads while protecting other customer classes is crucial.

Fourth, FERC should endeavor to utilize existing programs and market products that may achieve the same goals set out by the DOE to enhance flexibility of large loads, including demand response. These programs sit squarely within wholesale jurisdiction and may bring about the same benefits as “curtailment.” These existing programs may be

used to both provide a large load with firm service, as well as to hedge against curtailment risk during grid emergencies, as explained in the third principle.

Several RTOs have been considering how to address the curtailment of large loads. There are many ways FERC could navigate the need for curtailability. NRDC proposed one potential simple solution to curtailment in the PJM CIFP process: curtailments for any non-capacity backed large loads (i.e., large loads that have not paid for firm service) would be determined by PJM at the zonal level. For example, if a 500 MW new large load connects to PJM with 300 MW of new capacity (in the form of new supply or demand response deliverable to the new large load), the remaining 200 MW of non-firm load could be subject to curtailment. During an emergency, PJM would allocate this curtailment at the wholesale level, by telling the relevant Locational Deliverability Area that it needs to curtail load by 200 MW. It would be up to the state regulators to further allocate this curtailment between retail customers. This separation of responsibilities could help avoid jurisdictional disputes.

In the PJM CIFP, NRDC attempted to balance the risk of curtailment of large loads and the risk of high prices to other customers by proposing to curtail any large loads that have not secured firm service with other demand response resources on the early side, but always prior to scarcity pricing intervals. Additionally, any curtailment requirements must not allow a backup generator to be in violation of state or federal air quality laws or maximum backup generation run-time requirements. FERC could also provide guidance on pathways for large loads to hedge against curtailment risks, for example by utilizing energy-only resources to net against demand response.

Overall, large loads can pose serious risks to the public if guaranteed firm service *without* a plan for flexibility or to bring on new supply – indeed, “firm service” may not be a guarantee that RTOs can make for any customer based on current projections. We agree with the intent of encouraging flexibility in order to minimize reliability risks, but believe that the four principles outlined here must be respected while developing any curtailment or wholesale interconnection guidelines. To provide FERC with further information about the practicalities of curtailment, we recommend FERC host a technical conference and invite data center developers, NERC, and others to discuss these practicalities.

D. Network Upgrades, Transmission Costs, and Ancillary Services

1. DOE NOPR: Principles 8 through 12

Principles 8 through 12 generally aim to ensure that new large loads and co-located loads remain responsible for costs associated with their interconnection to the grid, and that they receive comparable treatment to other resources connecting to the grid.⁶¹

Principle 8 is that load and co-located loads should be fully responsible for the network upgrades associated with their interconnection, and FERC and DOE specifically seek comment on whether network upgrade costs can be offset through a crediting mechanism over time.

⁶¹ DOE ANOPR at PP 25-29.

Principle 9 is that load and co-located loads should have an equivalent option to build network upgrades as interconnecting generators.

Principle 10 requires that when generating facilities seek to serve new large loads rather than meet existing demand, a study must evaluate any network upgrades necessary to maintain reliability. Principle 10 requires that the generator fund network upgrades that the study finds necessary, and that the generator may suspend service to existing demand only after the upgrades are complete. The ANOPR seeks comment on whether the study should address resource adequacy.

Principle 11 requires utilities serving load and co-located loads to be responsible for transmission service based on the facilities' withdrawal rights.

Principle 12 requires utilities serving large load and co-located loads to be responsible for ancillary services based on peak demand.

2. Comments

These principles are generally consistent with the rule that costs must be allocated roughly commensurate with benefits and will help to protect against unreasonably shifting costs. As such, PIOs broadly support these principles, subject to the comments below.

As FERC crafts a rule regarding the interconnection of large or co-located loads, the FPA's fundamental cost-allocation precepts should be the guiding star. Under those precepts, "all rates must reflect to some degree the costs actually caused by the customer

who must pay them.”⁶² Where an entity “benefits from the costs of new facilities, it may be said to have ‘caused’ a part of the facilities.”⁶³ This principle prevents shifting costs to ratepayers who “derive no benefits, or benefits that are trivial in relation to the costs sought to be shifted.”⁶⁴

Under these fundamental cost-allocation principles, FERC’s rule must ensure the just and reasonable allocation of the costs of network upgrades and transmission costs. Any rule that FERC issues in this docket must ensure that new large loads or co-located loads bear responsibility for the costs they cause by interconnecting to the grid. Similarly, the rule must prevent the shifting of costs caused by large load interconnections onto other consumers who receive no benefits, or trivial benefits, from the network upgrades required to serve the interconnecting large loads. To the extent that current tariffs lack mechanisms to identify the beneficiaries of network upgrades, FERC should require updates to existing tariffs to guarantee that costs can be allocated in a just and reasonable manner.

a. Comments on Principle 8

Principle 8—that “load and [co-located loads] should be responsible for 100% of the network upgrades that they are assigned through the interconnection studies”—is generally consistent with the FPA’s cost-allocation principles. Where large load interconnections require network upgrades, the large loads actually cause the costs to be incurred and must be responsible for those costs. Moreover, new large loads or co-located

⁶² *Ill. Com. Comm’n v. FERC*, 576 F.3d 470, 476 (7th Cir. 2009).

⁶³ *Id.*

⁶⁴ *Id.*

loads are the sole beneficiaries of the network upgrades required for their interconnection; neither the new large loads nor the network upgrades required to serve them provide significant benefits to the broader grid or other ratepayers. For example, data from the PJM region show that transmission upgrades driven by data center loads have not led to system benefits, but rather only provide the benefit of allowing the large load to operate.⁶⁵

Project / Location	Scope / Cost	Purpose / Trigger	Why It Doesn't Benefit the Whole System	Planning Process	Who Pays	Source
Wishing Star Substation – Loudoun County, VA	\$627.6M – New 500/230 kV substation with tie-ins to Mosby and Brambleton lines	7.5 GW of projected load growth in 'Data Center Alley'	Tailored to serve hyperscaler reliability needs; no evidence of CETL or transfer capacity benefit	RTEP – 2022 Window 3	Dominion Zone	2022 PJM RTEP Report, p. 61
Marley Neck 115 kV Substation – BGE Zone	\$107.6M – Substation upgrade with 230/115 kV transformers	Overloads at seven Baltimore-area facilities	PJM explicitly states 'no broader system benefits'	RTEP – 2024 Window 1 (Baseline Project b3906)	BGE Zone (Baseline)	2024 RTEP Window 1 Report, Proposal 295
Beaumeade STATCOM – Northern Virginia	300 MVAR Static Synchronous Compensator at Beaumeade Substation	Reactive power support for clustered datacenter load	Voltage-only fix; does not improve bulk power flow or transfer capability	RTEP – 2023 Baseline	Dominion Zone	2023 PJM TEAC Whitepaper, p. 12
Kraken Loop – Fredericksburg Corridor	\$704M – 500 kV loop and new Kraken substation; 230 kV upgrades deferred	Local overloads and data center-driven load growth near North Anna	Local components are deferred to M-3 process; no broader regional impact	RTEP – 2024 Window 3 (Revised Proposal 967)	Dominion (for now); 230 kV deferred to M-3	2024 RTEP Window 3 Summary, Proposal 967
Talen–Amazon Co-Located Load – Susquehanna Nuclear	Up to 960 MW behind-the-meter load; capped at 480 MW to avoid violations	Co-located load increase at nuclear plant	PJM confirmed upgrades would be triggered above 480 MW, with no broader system benefit	FERC Interconnection Filing – ER24-2172	Unknown (but customer would trigger upgrades >480 MW)	FERC Order ER24-2172

Because other ratepayers receive no benefits, or trivial benefits, from the network upgrades required to interconnect new large loads or co-located loads, it would be

⁶⁵ See Comments of Public Interest Organizations, Docket No. EL25-49, at 13 (Apr. 23, 2025), Accession No. 20250423-5243.

inconsistent with cost-allocation principles to allow the costs of those network upgrades to be shifted to other ratepayers. Hence, it is appropriate for the new large loads or co-located loads to bear 100% of the cost of the network upgrades required for their interconnection.

To effectuate this principle, FERC must ensure that load interconnection studies are sufficiently rigorous and transparent to identify the beneficiaries of network upgrades and enable cost-allocation that is roughly commensurate with the benefits. For example, current processes in the PJM region fail to rigorously assess the costs and beneficiaries of network upgrades to serve new large loads.⁶⁶ FERC needs to ensure that such processes are sufficiently rigorous and transparent.

FERC should not adopt a default crediting mechanism to offset the costs of network upgrades required for large load interconnection. Again, because large loads are the sole or primary beneficiaries of the network upgrades required for their interconnection, it is appropriate for them to bear 100% of the costs of those upgrades. To the extent that particular network upgrades could theoretically provide broader benefits to other ratepayers, the relevant transmission-owning utility should bear the burden of proposing a specific cost-allocation variation for that particular facility. To make such an individualized proposal, the relevant transmission-owning utility should be required to prove the degree to which the network upgrade would provide benefits to other ratepayers, and to specify what group of ratepayers would receive those benefits. In such

⁶⁶ See Comments of Public Interest Organizations, Docket No. EL25-49, at 7–24 (Apr. 23, 2025), Accession No. 20250423-5243 (explaining how PJM’s “necessary study process” and “do-no-harm” studies fail to gather sufficient information to enable a just and reasonable allocation of costs).

limited circumstances, a crediting mechanism might be a just and reasonable component of a facility-specific cost-allocation proposal.

b. Comments on Principle 9

Principle 9, which would provide large loads and co-located loads an option to build network upgrades equivalent to the option held by interconnecting generators, appears just and reasonable and consistent with cost-allocation principles to the same degree as equivalent provisions governing generator interconnection. Extending an equivalent option to build for interconnecting large loads and co-located loads could also help avoid arguments that FERC's rule treats loads in an unduly discriminatory manner.⁶⁷ Ultimately, this principle does not appear to shift costs of large load interconnections in a manner inconsistent with cost-allocation principles.

c. Comments on Principle 10

PIOs do not believe that the Commission should adopt Principle 10. Many regions of the country are currently experiencing resource adequacy or reliability challenges. Generation that has been on the system and serving existing load should not be allowed to be essentially removed from the grid to serve a single customer. In many cases, costs of these facilities – or the network upgrades needed to interconnect them initially – have been paid for by existing customers. FERC should not allow any single customer to then claim either full or partial rights to the existing generation. Principle 10 considers the

⁶⁷ Such arguments may lack merit, because interconnecting large loads are not entirely similarly situated to interconnecting generators. Interconnecting generators provide significant benefits to the grid, including by providing resource adequacy and competition in the wholesale markets, whereas interconnecting large loads do not. Nevertheless, there may be value to avoiding disputes about purportedly undue discrimination.

circumstance in which “an existing generation facility seeks to enter a partial suspension to serve a new load at the same location.”⁶⁸ However, the suspension of existing generation to serve new large loads is inconsistent with cost-causation principles and unjust and unreasonable because this practice shifts costs associated with maintaining resource adequacy onto other consumers, as well as increasing reliability risks for existing ratepayers.

When an existing generator suspends its service to existing load in order to instead serve a new load, that suspension will increase costs associated with maintaining resource adequacy for other consumers. In regions without an organized capacity market, such a suspension may increase the frequency or intensity of scarcity pricing in energy markets. In regions with organized capacity markets, such a suspension reduces the available supply of capacity, which will in turn increase capacity costs. In either case, the effects of shifting these costs to existing consumers may be profound. For example, in PJM, where the capacity market is currently experiencing tight conditions, even seemingly small reductions in the supply of capacity can impose billions of dollars in increased costs.⁶⁹ While suspending service to the broader market to serve an interconnecting large load may be economically attractive for the particular generator and the new large load, the

⁶⁸ DOE ANOPR at P 27.

⁶⁹ See Md. Office of People’s Counsel, Bill and Rate Impacts of PJM’s 2025/2026 Capacity Market Results & Reliability Must-Run Units in Maryland, at 8 (Aug. 2024) (“Synapse Report”), https://opc.maryland.gov/Portals/0/Files/Publications/RMR%20Bill%20and%20Rates%20Impact%20Report_2024-08-14%20Final.pdf?ver=V9hZfyTmjLeNVt2Dg3cTgw%3d%3d; Monitoring Analytics, Analysis of the 2025/2026 RPM Base Residual Auction Part A, at 2 (Sept. 20, 2024) (“IMM Analysis of 2025/2026 Auction”), https://www.monitoringanalytics.com/reports/Reports/2024/IMM_Analysis_of_the_20252026_RPM_Base_Residual_Auction_Part_A_20240920.pdf (documenting how the exclusion of only roughly 500 MW of generation from the capacity market inflated capacity market clearing prices by between \$4 billion and \$5 billion).

resulting cost shifts to other consumers are unjust and unreasonable and inconsistent with cost-causation principles because other consumers receive no benefits from the generator's suspension or the large load's interconnection.

These problems may be especially acute in states where traditional utilities own generation. In those states, the inclusion of generation in the utilities' rate base means that other consumers have already paid for the construction and maintenance of the existing generation; hence, the removal of the benefits for which those consumers have already paid exacerbates the problem of increased costs of paying for resource adequacy moving forward.

Suspending an existing generator to serve new loads also unjustly and unreasonably imposes reliability risks on other consumers. Where existing generation is necessary to maintain resource adequacy, its suspension increases the possibility that supply will be insufficient to meet demand, potentially resulting in disruptions of electric service. It is unjust and unreasonable to force ordinary ratepayers to bear an increased risk of blackouts simply to facilitate new large loads buying up existing capacity.

Principle 10 only partially mitigates these concerns by requiring "a system support resource (SSR)/reliability must run (RMR) type study." Although allowing the suspension of existing generation to serve new large loads would not be just and reasonable, Principle 10 works to mitigate the reliability harms from that approach by: (1) requiring a study to assess a suspension's reliability impacts, (2) requiring the suspending generator to pay the full cost of any network upgrades needed to maintain reliability, and (3) allowing suspension only after those network upgrades are in place. That approach

mitigates the risk that a sudden suspension of existing generation could cause immediate reliability risks. However, that approach does not protect existing consumers against the increased costs associated with the existing generator's suspension.

Ultimately, Principle 10 could allow the suspension of generators in a manner that unjustly and unreasonably imposes additional costs and risks on existing consumers, and Principle 10 would only partially mitigate those adverse impacts. For these reasons, FERC should instead establish a default rule that does not permit the suspension of existing generation to serve new large loads. However, to the extent that FERC does permit the suspension of existing generation to serve new large loads, Principle 10 offers a good first step to mitigate adverse reliability impacts from this approach. FERC should also supplement Principle 10 by adopting additional measures to safeguard existing consumers against increased costs caused by the suspension of existing generation.

d. Comments on Principle 11

FERC should clarify Principle 11 to avoid ambiguity. As drafted, Principle 11 does not make entirely clear whether it intends to describe the responsibility of transmission-owning utilities, interconnecting large loads, or both. The intent appears to be to limit the amount of transmission service that transmission-owning utilities must provide, and that interconnecting large loads must purchase, based on the interconnecting large load or co-located loads' withdrawal rights. Hence, the principle appears to apply to both transmission-owning utilities and interconnecting large loads.

Under Principle 11, FERC can reasonably limit the transmission service that transmission-owning utilities must provide, and that interconnecting large loads or co-

located loads must purchase, only in concert with the ideas expressed in Principle 6.

Principle 6 requires co-located loads to “install the system protection facilities necessary to prevent unauthorized injections or withdrawals that exceed the respective rights.”⁷⁰

Similar system protection facilities should be required in order to limit the physical ability of new large loads or co-located loads to utilize more transmission service than their withdrawal or injection rights allow. Without such controls, Principle 11 could unreasonably limit the study and buildout of the transmission grid and risk reliability problems that could result if new large loads or co-located loads withdraw or inject more energy than their rights allow. However, if new large loads and co-located loads are properly controlled, then limiting the responsibility of transmission-owning utilities to provide transmission service, and limiting the responsibility of new large loads or co-located loads to purchase transmission service, based on the new large load or co-located loads’ withdrawal rights may be appropriate.

e. Comments on Principle 12

FERC should also clarify Principle 12. As currently drafted, Principle 12 states that “utilities serving large loads, including those at [co-located loads], should be responsible for ancillary services.”⁷¹ FERC should make clear that this principle governs utilities’ responsibility to procure ancillary services. In particular, this principle should clarify that utilities’ procurement of ancillary services should be based on peak demand

⁷⁰ DOE ANOPR at P 23.

⁷¹ DOE ANOPR at P 29.

rather than being reduced by the degree to which co-located generation may reduce net peak demand by offsetting some of the demand of a new large load.

We tentatively agree with the principle that utilities' procurement of ancillary services should be based on total peak demand, and not reduced by expected contributions from generation co-located with large loads. In general, we believe that calculating utilities' needs for ancillary services based on peak demand should be more accurate than reducing the need for ancillary services based on the expected performance of co-located generation. Similarly, to the extent co-located generation does not reduce the overall need for utilities to procure ancillary services (meaning that co-located generation's potential contributions of ancillary services are not accounted for through such a reduction), it appears reasonable to allow those co-located generation resources to offer ancillary services into the relevant markets.

However, our support for Principle 12 is tentative for two reasons. First, co-located generation may have limited ability to provide ancillary services, due to limits imposed by control technologies that limit energy withdrawals or injections, or due to contractual limits on withdrawals or injections.

More fundamentally, calculating utilities' requirements to procure ancillary services based on peak demand could result in shifting the costs of ancillary services caused by the addition of large loads to existing consumers. To the extent that new large loads increase the overall demand for ancillary services, it is essential that the new large loads pay their fair share for such services. Ancillary services, including blackstart services and others, are crucial to the reliable operation of new large loads. And because

even co-located loads will rely on such ancillary services, it is essential that they pay their fair share.⁷² FERC should evaluate what mechanisms within its jurisdiction can ensure that new large loads pay their fair share for ancillary services commensurate with the degree to which new large loads cause the need for such services. To the extent FERC concludes that the allocation of the increased cost of ancillary services is not within its jurisdiction, but is instead a matter that states must regulate as part of retail rates, FERC should at least ensure that the rules within FERC's jurisdiction provide as much transparency as possible regarding the degree to which different types of load contribute to the need for ancillary services, so that states will be well-equipped with the necessary information to ensure just and reasonable cost allocation at the retail level.

E. Reliability Reviews

1. DOE NOPR: Principle 14

Principle 14 would require utilities serving large loads to meet all applicable NERC reliability standards and Open Access Transmission Tariff (OATT) provisions. It also states that NERC should review its reliability standards to determine if new registration categories or new or modified reliability standards are required to ensure reliability of the bulk-power system.

2. Comments

⁷² See, e.g., Comments of the Union of Concerned Scientists, Docket No. EL25-49-000, (Apr. 10, 2025), Accession No. 20250411-5520.

Jim Robb, President and Chief Executive Officer of NERC, said it best: “There is existing evidence that large loads impact BPS reliability.”⁷³ Any FERC rulemaking on this issue must ensure reliability. Thus, we agree with the principle that utilities must continue to meet mandatory reliability standards. We also agree that NERC should take a look at its reliability standards and registration to determine if changes are warranted. However, we note that the DOE Proposal missed one critical principle on reliability – it does not make clear that large loads must also meet any applicable reliability standards that may apply to them in the future. FERC should be sure to remedy this oversight.

As noted above, section 215 of the FPA gives FERC jurisdiction over large loads as users of the bulk-power system.⁷⁴ NERC has set up a Large Load Task Force “to better understand the reliability impact(s) of emerging large loads ... and their impact on the bulk power system.”⁷⁵ Mr. Robb has indicated that in the first quarter of 2026, NERC will develop additional findings, guidance, and steps forward to account for the BPS reliability related needs associated with large load.

While NERC has indicated that it may need to “enhance” its existing reliability standards, it should also assess whether it should register large loads as users of the bulk-power system and develop new reliability standards. NERC and its Large Load Task Force are a useful forum to do this work, but it should ensure that it moves swiftly. If

⁷³ Pre-Conference Comments of the North American Electric Reliability Corporation, 2025 Reliability Technical Conference, Docket No. AD25-8-000 at 15 (October 21, 2025).

⁷⁴ See *infra* section c. See also 16 U.S.C. § 824o(b)(1).

⁷⁵ NERC Large Loads Task Force (LLTF), <https://www.nerc.com/comm/RSTC/Pages/LLTF.aspx>.

necessary, FERC could order NERC to submit new or modified reliability standards to FERC to address large loads.⁷⁶

F. Implementation Plans

1. DOE NOPR: Principle 13

Under Principle 13 there must be a plan to implement any reforms that FERC adopts. DOE seeks comment on appropriate transition plans, including the treatment of large load interconnections that are already being studied for interconnection.

2. Comments

FERC should take care to adopt rules that do not disrupt existing processes – either generator or large load interconnections. We agree that FERC must create clear guidance for large load interconnections already under interconnection assessment and those that will soon join the study queue. FERC needs to clarify how far along in the interconnection study process a new large load must be in order to be exempt from FERC’s new standards, and/or if existing projects will need to be restudied to comply with FERC’s new standards. FERC also needs to clarify in its transition plans how long of a runway utilities will have for compliance so that they can anticipate which portfolio of interconnection requests will need to apply any new FERC standards. Finally, FERC should clarify in its transition plans whether and how FERC will require clusters to be formed for interconnection study.

G. Other Issues

1. Comments

⁷⁶ 16 U.S.C. § 824o(d)(5).

FERC may need to consider issues not addressed on the DOE ANOPR. For example, the DOE Proposal does not discuss how FERC's anti-market manipulation authority related to the purchase or sale or transmission of electricity under Section 222 of the FPA might apply to large loads. For example, co-location arrangements involving existing generators can heighten the concern with deficiencies in market power mitigation rules by exacerbating certain market participants' incentives to abuse market power and engage in market manipulation. A market participant can exercise market power or engage in market manipulation by withholding supply.⁷⁷ The incentive to withhold supply in energy markets grows as the portions of the supply curve used to meet demand are more inelastic.⁷⁸ This is because a given amount of withheld supply will tend to have a greater impact on price as that portion of the supply curve becomes more inelastic. Co-location arrangements that allow load to co-locate with existing generators have the potential to exacerbate the incentive to withhold supply by changing the generators on which the market relies to meet demand. If existing generators' co-location causes the market operator to more frequently rely on more inelastic portions of the supply curve to meet demand, then market participants' incentive to withhold supply is greater.⁷⁹

Indeed, FERC has already had to exercise elements of this authority in various proceedings involving large load customers.⁸⁰ In particular, large loads have access to

⁷⁷ See, e.g., *Pub. Citizen v. FERC*, 7 F.4th 1177, 1198 (D.C. Cir. 2021); *FPA Section 203 Suppl. Pol'y Statement*, 120 FERC ¶ 61,060, at PP 60, 65 (2007).

⁷⁸ See, e.g., *Commonwealth Edison Co.*, 91 FERC ¶ 61,036, at 61,133 & 61,133 n.42 (2000).

⁷⁹ See also PIOs Comments (Docket No. EL25-49-000) at 32-33.

⁸⁰ See FERC Docket Nos. ER24-2172-000 / ER24-2172-001.

behind-the-meter generation, which they may be able to use to manipulate market pricing to their advantage. FERC must determine whether changes to its market power and/or market manipulation rules – or the tariff provisions dealing with market power – are necessary to provide the guardrails needed to prevent abuses of market power or market manipulation.⁸¹

IV. Conclusion

We appreciate the opportunity to comment on the principles outlined in the DOE ANOPR. The DOE Proposal is a good start at setting principles for how FERC can regulate the interconnection of large loads, but there are many details to be worked out. In doing so, FERC must follow the guiding principle that new large loads should not be permitted to harm affordability or reliability for existing consumers. New large loads should be responsible for the costs they cause, they should not be allowed to imperil reliability for other consumers, and FERC cannot allow utilities to unduly preference large load customers over other types of customers. We look forward to seeing FERC's proposal on these matters.

⁸¹ See *Practical Guidance and Considerations for Large Load Interconnections* at 31 Gridlab (May 2025).

Respectfully submitted this 21st day of November, 2025,

/s/ Christy Walsh
Christy Walsh
Senior Attorney and Director of Federal
Energy Markets
Sustainable FERC Project
1152 15th St NW #300
Washington, DC 20005
202-909-6842
cwalsh@nrdc.org

/s/ Jackson Morris
Jackson Morris
Director, State Power Sector
Natural Resources Defense Council
40 W 20th St 11th Floor
New York, NY 10011
212-727-4468
jmorris@nrdc.org

/s/ Danielle C. Fidler
Danielle Fidler, Senior Attorney
Clean Air Task Force, Inc.
114 State Street, 6th Floor
Boston MA 02109
Dfidler@catf.us

/s/ Elizabeth R. Marx, Esquire
/s/Patrick M. Cicero
Elizabeth R. Marx, Esquire
Patrick M. Cicero, Esquire
*Counsel for Coalition for Affordable
Utility Services and Energy Efficiency in
Pennsylvania*
Pennsylvania Utility Law Project
118 Locust Street
Harrisburg, PA 17101
717-236-9486, Ext. 204
emarx@pautilitylawproject.org
pcicero@pautilitylawproject.org

/s/ Nick Lawton
Nick Lawton, Senior Attorney, Clean
Energy Program
Christine Powell, Deputy Managing
Attorney, Clean Energy Program (FERC)
Aaliyah Adkins, Associate Attorney, Clean
Energy Program
Earthjustice
1001 G Street, NW Suite 1000
Washington, DC 20001
202-780-4835
nlawton@earthjustice.org
cpowell@earthjustice.org
aadkins@earthjustice.org

/s/ Ted Kelly
Ted Kelly
Director and Lead Counsel, US Clean
Energy
Veronica Ung-Kono Fellow, US Clean
Energy
Environmental Defense Fund
555 12th Street NW, Ste 400 Washington,
DC 20004
tekelly@edf.org
vungkono@edf.org

/s/Nick Fitzmaurice
Nick Fitzmaurice
Energy Transition Engineer
Montana Environmental Information
Center
PO Box 1184, Helena, MT 59624
406-443-2520 x 007
nfitzmaurice@meic.org

/s/Gregory E. Wannier
Gregory E. Wannier, Senior Attorney
Justin Vickers, Senior Attorney
Sierra Club Environmental Law Program
2101 Webster St., Suite 1300
Oakland, CA 94612
415-977-5646
greg.wannier@sierraclub.org
justin.vickers@sierraclub.org

/s/Nick Guidi
Nick Guidi
Senior Attorney
Southern Environmental Law Center
122 C Street NW, Suite 325
Washington, DC 20001
202-573-8136
nguidi@selc.org

CERTIFICATE OF SERVICE

I hereby certify that the foregoing has been served in accordance with 18 C.F.R. § 385.2010 upon each party designated in the official service list compiled by the Secretary in this proceeding, by email.

/s/ Christy Walsh
Senior Attorney and Director of Federal
Energy Markets
Natural Resources Defense Council,
Sustainable FERC Project
1152 15th St NW #300
Washington, DC 20005
202-909-6842
cwalsh@nrdc.org